

Profile

I'm a Singaporean developer building simulations, tools, and interpretability research, most of it hosted at a9l.im. Literally everything I make is written with Claude: vanilla JS for the site, plus Python and Rust for research. I'm open to freelance and collaboration, especially research tools and anything DIY.

Experience

- Independent Developer – a9l.im**

Feb 2026 – Present

 - Built and maintain, solo, a portfolio of open-source interactive simulations and tools plus the shared design system and component library used across all of them, all in rawdogged vanilla JavaScript.
 - Architected the SSR layer on Cloudflare Workers and Assets: per-route HTMLRewriter injection, edge-rendered markdown, structured data, and per-route security headers.
- SDDM Theme Maintainer – Catppuccin**

2025 – Present

 - Led the QtQuick rewrite of Catppuccin's SDDM display-manager theme: dynamic accent colors, per-user icons, designed vector backgrounds, and automated theme generation across all four flavors.

Research

- Saklas**

PyPI · Python

 - Activation steering and probing library powering the studies below: pulls contrastive steering vectors (Zou et al. 2023) and adds them to hidden states at generation, no fine-tuning. Ships 26 in-flight probes for affect, stance, register, and alignment.
 - Python API, an OpenAI- and Ollama-compatible server, and a live terminal UI; supports 55 model architectures. On PyPI under AGPL-3.0 with CI and llama.cpp GGUF interchange.
- Introspection via Kaomoji**

Python · HuggingFace

 - Built llmoji, a CLI that hooks coding agents (Claude Code, Codex, Hermes) to open every turn with a kaomoji, then synthesizes a meaning per face and ships the corpus as a public HuggingFace dataset.
 - Found that a model's hidden-state affect quadrant is recoverable from its kaomoji choice across open-weight families (by Jensen-Shannon similarity, not argmax). Sibling studies probe a residual-stream attractor basin and whether stated elapsed time is represented or confabulated.

Selected Projects

- Ogdoad**

Rust · Python

 - Clifford algebras and quadratic-form classification over the commutative number systems adjacent to Conway's combinatorial games, laid out as one table by place (Archimedean, p-adic, transfinite, finite) and characteristic. Surreals recover the real Cl(p,q) classification with infinite and infinitesimal metric entries; characteristic-2 legs like the nimbers swap the 8-fold periodicity for the Arf invariant and Brauer–Wall group. Verified Rust core with Python bindings.
- Bio-transformer**

Python

 - Neuromorphic transformer from three brain subsystems that each compute something transformer-shaped: astrocytes as recurrent linear attention, the Marr–Albus cerebellum as the FFN, and laminar neocortex as the residual stream via predictive coding. Every neuron is a leaky integrate-and-fire spiker and every weight update is local (Oja, Hebbian, and climbing-fibre rules), with no backprop or surrogate gradient.
- Interactive Simulations – a9l.im**

JavaScript · WebGPU

 - Geon**: relativistic N-body simulator on WebGPU compute shaders, with 11 forces, Barnes-Hut scaling, and Kerr-Newman black holes. **Shoals**: options-trading sim with Heston, Merton, and Vasicek pricing and a 400+ scenario event engine.
 - Plus **Plasma** (WebGPU resistive MHD), **Pile** (PWR, RBMK, and molten-salt reactor neutronics), **Miasma** (stochastic spatial epidemics), **Cyano** (cellular metabolism), **Gerry** (redistricting and fairness metrics), and **Scripture** (sixteen-text reader with full-text search and edge SSR).

Education

- University of California, San Diego**

March 2026

B.S. in Mathematics · GPA 3.75 · GRE 335 (170Q, 165V)
- Singapore American School**

Class of 2023

Summa Cum Laude · GPA 4.50

Skills

AI & interpretability: Claude Code and Codex daily, integrating AI-generated code at scale; activation steering, contrastive-PCA probing, and hidden-state analysis on HuggingFace transformers; MCP server authoring.
Languages: JavaScript (Canvas, WebGL, WebGPU, GLSL, WGSL), Python (PyTorch, HuggingFace transformers, NumPy), Rust, Java, QtQuick and QML, LaTeX.
Web & infrastructure: Cloudflare Workers and Assets, edge SSR via HTMLRewriter, JSON-LD structured data, self-hosted typography, no-build pipelines.

Other: Technical writing, vector graphics, soldering, Spanish (novice), and conlang construction.